

Higher-Order Metaphysics
 Week Seven: Classicism and questions of grain
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1 The terrain between Classicism and Extensionalism

$$\begin{aligned}
 \text{LB}_\tau &:= \lambda y^\tau X^{\tau \rightarrow t}. \forall z^\tau. Xz \rightarrow y \leq_\tau z \\
 \text{GLB}_\tau &:= \lambda y^\tau X^{\tau \rightarrow t}. \forall z^\tau. \text{LB}_\tau zX \leftrightarrow z \leq_\tau y \\
 \text{Atom}_\tau &:= \lambda y. \forall z. (z \leq_\tau y \wedge z \neq y) \leftrightarrow z \leq_\tau \neg_\tau z \\
 \text{Persistent} &:= \lambda Y. \Box \forall \vec{z}. Y\vec{z} \rightarrow \Box Y\vec{z} \\
 \text{Inextensible} &:= \lambda Y. \Box \forall X. \forall \vec{z}. Y\vec{z} \rightarrow \Box X\vec{z} \rightarrow Y \leq X \\
 \text{Rigid}_{\sigma_1, \dots, \sigma_n} &:= \lambda Y. (\Box \forall X. (\forall \vec{z}. Y\vec{z} \rightarrow \Box X\vec{z}) \leftrightarrow Y \leq X) \\
 \text{Functional}_{\vec{\sigma}, \tau} &:= \lambda U^{\vec{\sigma} \rightarrow \tau \rightarrow t}. \forall x^{\vec{\sigma}} \exists y^\tau. Uxy \wedge \forall z (Uxz \rightarrow y = z)
 \end{aligned}$$

$$\begin{aligned}
 \text{Boolean Completeness} & \quad \forall X^{\tau \rightarrow t} \exists y^\tau. \text{GLB}_\tau yX \\
 \text{Atomicity} & \quad \forall x. x \leq \neg_\tau x \vee \exists y. \text{Atom}_\tau y \wedge y \leq_\tau x \\
 \text{Actuality} & \quad \exists p. p \wedge \forall q (q \rightarrow p \leq q) \\
 \text{Rigid Comprehension} & \quad \forall X. \exists Y. \text{Rigid}(Y) \wedge \forall \vec{z}. X\vec{z} \leftrightarrow Y\vec{z} \\
 \text{Functional Plenitude} & \quad \forall R^{\sigma \rightarrow \tau \rightarrow t}. \text{Functional}(R) \rightarrow \exists X^{\sigma \rightarrow \tau}. \forall y. Ry(Xy)
 \end{aligned}$$

2 Maximalizing a theory

Suppose T is a theory in a higher-order language $\mathcal{L}$. Then the *maximalization* of T , $\text{Max } T$, is the smallest extension of T containing all **closed** distinctness claims $A \neq_\sigma B$ which are consistent with T .¹

We could also define the *quantified maximalization* of T , $\text{Max}_q T$, as the smallest extension of T containing all closed sentences $\forall \vec{v}. A \neq_\sigma B$ which are consistent with T .²

3 Maximalizing an extension of Classicism

When T includes Classicism, we can give some other helpful characterizations of $\text{Max } T$.

Let $\text{Max}_\Diamond T$ be the smallest extension of T that includes $\Diamond P$ for every closed P consistent with T .

¹ We can either theories with sets of formulae closed under MP and Gen, or with sets of sentences closed under MP—I don't think it will matter.

² NB this isn't the same as adding all open formulae $A \neq_\sigma B$ consistent with T : on the normal way of talking about consistency for open formulae, $p \neq_t q$ is consistent with any T , but $\forall p^t q^t. p \neq_t q$ is inconsistent.

- Since whenever P is consistent with T , $P \neq \perp$ is too, $\text{Max } T$ will include $P \neq_t \perp$, i.e. $\Diamond_{\top} P$, for each such P , so $\text{Max}_{\Diamond} T \subseteq \text{Max } T$.
- Since $\forall \vec{v}. A \neq B$ entails $\Diamond \forall \vec{v}. A \neq B$, $\text{Max}_{\Diamond} T$ includes $\Diamond \forall \vec{v}. A \neq B$ whenever $\forall \vec{v}. A \neq B$ is closed and consistent with T ; but in Classicism, $\Diamond \forall \vec{v}. A \neq B$ entails $\forall \vec{v}. \Diamond(A \neq B)$ (by CBF) and hence $\forall \vec{v}. (A \neq B)$ (by NI). So $\text{Max}_q T \subseteq \text{Max}_{\Diamond} T$.

Since trivially $\text{Max } T \subseteq \text{Max}_q T$, we have that for T extending Classicism, $\text{Max } T = \text{Max}_q T = \text{Max}_{\Diamond} T$.³

³ This reasoning will also go through in “L-Classicism” (see last week’s handout) and other weakenings of Classicism in a similar spirit, with $\Diamond_{\top}$ redefined as $\lambda p. \neg(p \equiv_L \perp)$.

4 “At least as coarse-grained as”

Intuitively, Classicism is a rather “coarse-grained” theory of higher-order reality, though not as coarse-grained as Extensionalism. What does “coarse-grained” mean in this sort of setting?

Bacon and Dorr (forthcoming: 2.4) suggest the following operational definition: T_1 is at least as coarse-grained as T_2 (T_2 is at least as fine-grained as T_1) iff (i) every closed identity in T_2 is in T_1 , and (ii) every closed distinctness claim in T_1 is in T_2 .⁴

By this definition, $\text{Max } T$, if consistent, is at least as fine-grained as every consistent extension of T .⁵

A more discriminating definition (maybe a bit better) would replace closed identities and distinctness claims with *universally quantified* identities and distinctness claims. On this definition, $\text{Max}_q T$, if consistent, will be at least as fine grained as any consistent extension of T .

⁴ Note that neither this definition nor anything else in a similar spirit will give a total ordering!

⁵ Clause (ii) is automatic. And so long as $\text{Max } T$ is consistent, it can’t prove any closed identities that T doesn’t already prove, since if $A = B$ were such an identity, $\text{Max } T$ would include $A \neq B$ as well as $A = B$ and thus be inconsistent.

5 Which theories have consistent maximalizations?

We can show that the maximalizations of Extensionalism, C5, and $C + B$ are inconsistent. All these theories are consistent with $(\Box_{\top} \exists x^e y^e. x \neq y)$ and its negation, but not with $\Diamond(\Box_{\top} \exists x^e y^e. x \neq y) \wedge \Diamond \neg(\Box_{\top} \exists x^e y^e. x \neq y)$.

Fritz, Lederman, and Uzquiano (2021) proves that $\text{Max } H$ and $\text{Max } H_0$ are consistent.⁶

Bacon and Dorr (forthcoming: appendix E) proves that $\text{Max } C$ is consistent.

Goodsell (***) proves that $\text{Max } T$ is inconsistent when T is any extension of Classicism that includes Rigid Comprehension.

⁶ Are $\text{Max}_q H$ and $\text{Max}_q H_0$ consistent? An interesting question I haven’t investigated—could be easy!

6 Computability issues

For many theories including that of extensions of Classicism⁷, we can show that $\text{Max } T$ is not *recursively enumerable*, and hence not *recursively axiomatizable*.⁸ This arguably means that humans aren't in a position to ""adopt"" such theories in the same sense in which we might adopt r.e. theories.

This leads to an interesting research program of formulating strong but simple r.e. fragments of interesting maximalized theories.⁹

⁷ wwhich others?

⁸ Note that this depends on $\text{Max } T$ being consistent, since the inconsistent theory in any signature is decidable.

⁹ See 'Classicism', footnote 53.

7 Why many plausible theories' maximalizations are not plausible

If T is a logic (theory closed under substitution of arbitrary expressions for nonlogical constants), and its language has nonlogical constants corresponding to groups of English words like ('vixen', 'female', 'fox') or ('tadpole', 'juvenile', and 'frog'), $\text{Max } T$ looks very implausible! For in such a language, identities like $\text{tadpole} = (\lambda x. \text{juvenile } x \wedge \text{frog } x)$ look very plausible, but $\text{Max } T$ will contain their negations.

(Although I find it implausible, a view that just rejects all such higher-order identities is still interesting to think about. The challenge it faces is to explain what *stops* it from sometimes happening that simple expression denotes something already denoted by some complex expression.)

This sort of objection may be available even if there's only one random nonlogical constant.

But for certain special signatures of nonlogical constants, this sort of objection to maximalized theories is not available.

- EG: A signature for which it would be reasonable (for someone comfortable with the ideology of 'fundamentality') to think that all the constants denote distinct *fundamental* properties/relations/individuals...
- EG: a signature with no nonlogical constants at all!

Interesting Fact: if T is a logic and closed under ζ ($Fx = Gx/F = G$), $\text{Max } T$ is $\text{Max } T^- + \text{Separated Structure}$, where T^- is the set of theorems of T that don't have any nonlogical constants, and Separated Structure is

Separated Structure $Fc = Gc \rightarrow F = G$, where c is a non-logical constant and F and G are closed terms not including c .

8 Maximalized theories and the avoidance of “arbitrariness”

9 Maximalized theories and “recombination” principles

10 Fundamental Possibility

For those who are comfortable talking about fundamentality (for properties/relations) at all, it is natural to give ourselves a way of doing so in the formal language, by adding a family of nonlogical constants $\mathbf{Fun}_\sigma$.

Using this, we can consider a schema whose attractions do not depend on those of any *specific* hypothesis about what is fundamental:

Fundamental Possibility $\mathbf{Fun} \vec{x} \rightarrow \Diamond P$, where P is any formula consistent with T with no nonlogical constants¹⁰ and whose free variables are among $\vec{x}$.¹¹

¹⁰ Not even $\mathbf{Fun}_\sigma$!

¹¹ A worry about this: it entails that no relation and its converse are both fundamental. We can avoid this by replacing $\neq$ in the definition of $\mathbf{Fun}$ with ‘is not a generalized converse of’.

11 Beyond maximalization

The kind of thinking that might make theories like Maximalist Classicism seem attractive seems to suggest further thoughts that do not follow from Maximalist Classicism. Two extensions discussed in the paper:

Possibility+ $(\bigwedge_{1 \leq i < j \leq n} x_i \neq x_j) \rightarrow \Diamond P$

where P has only free variables $\vec{x}$, all of type e and is consistent in (Classicism).

Strong Possibility $\Diamond_{\neq} P$, where P is closed and consistent with *Maximalist* Classicism.

References

- Bacon, Andrew, and Dorr, Cian (forthcoming), “Classicism”, in Peter Fritz and Nicholas K. Jones (eds.), *Higher-Order Metaphysics* (Oxford: Oxford University Press).
- Fritz, Peter, Lederman, Harvey, and Uzquiano, Gabriel (2021), “Closed Structure”, *Journal of Philosophical Logic*, 50: 1249–91.